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A dynamic VM consolidation technique for QoS and energy consumption in cloud environment

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Abstract Cloud-based data centers consume a significant amount of energy which is a costly procedure. Virtualization technology, which can be regarded as the first step in the cloud by offering benefits like the virtual machine and live migration, is trying to overcome this problem. Virtual machines host workload, and because of the variability of workload, virtual machines consolidation is an effective technique to minimize the total number of active servers and unnecessary migrations and consequently improves energy consumption. Effective virtual machine placement and migration techniques act as a key issue to optimize the consolidation process. In this paper, we present a novel virtual machine consolidation technique to achieve energy–QoS–temperature balance in the cloud data center. We simulated our proposed technique in CloudSim simulation. Results of evaluation certify that physical machine temperature, SLA, and migration technique together control the energy consumption and QoS in a cloud data center.

Keywords Cloud computing · Resource management · Server consolidation · Energy consumption

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1 Introduction

Cloud computing is a type of computing that delivers hosted services such as VoIP (Google Voice), social applications (Facebook), media services (YouTube), email (Gmail), banking and financial services, and much more over the Internet through a browser or web interface. Cloud computing enables individuals and companies to consume data, compute, storage, and application services just like electricity as they need. Data center host cloud computing is known as Cloud Data Center. According to the US National Institute of Standards and Technology, "cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction [1]. Cloud computing has made computing as a utility, new features and capabilities for individuals and companies such as self-service provisioning, elasticity, pay as you go, and unlimited resource gives the potentiality to transform their IT infrastructure even more useful and purposeful as a service, platform, or infrastructure in the cloud [2]. Cloud providers, such as Amazon, Microsoft, Google, and IBM, are planning to develop new data center across the world for their users. A number of cloud data centers are growing fast, and the operation expenses increase due to the increasing energy consumption and environmental pollutants of the data center [3]. Cloud companies like Microsoft and Google solely have more than 1 million servers in their infrastructures based on existing reports on [4]. On the other hand, cloud providers want to get maximum revenue per infrastructure equipment and increase profit by preventing paying due to the exorbitant cost of energy consumption using new equipment, new strategy, and new management tools. Recent research in the USA indicates that in 2013, 12 million computer servers in about 3 million data centers providing all the online service for the USA and energy consumption of these data centers are enough for 2 years of power consumption of New York city's households [5].

Recent research in IEEE [6] shows that data centers consume around 3 percent of the world's total electricity, producing 200 million metric tons of carbon dioxide. Data centers in the USA are predicted to consume 140 billion kilowatts per year by 2020, resulting in $13 billion per year in electricity bills. As a result, energy management has become one of the challenging research issues in cloud computing paradigm.

Virtualization is the fundamental technology that makes cloud computing possible and helps deliver cloud services. Virtualization allows host resources to be shared to create and run one and more virtual machines in a way that each virtual machine think it has their own processor, memory, bandwidth and other resources. It also gives the ability to efficiently manage the ever-increasing demand for storage, computing, and networking resources.

Virtualization can change modes of virtual machines and servers into standby to reduce infrastructure costs while increasing the efficiency without reducing system performance [7]. One of the beneficial aspects of virtualization technologies is virtual machines consolidation that allows us to consolidate services and application in a fewer resource; on the cloud, resource consolidation is a key technique in the infrastructure as a service layer and aims to improve optimization of equipment. Resource consolidation techniques should be optimized to avoid violating service-level agreements, performance degradation, resource fragmentation, and over-provisioning of resources emerges in cloud data centers. One way to reduce energy consumption is virtual machine consolidation in which virtual machines are periodically reallocated to minimize the number of servers by using live migration while still preserving application performance requirements and are delivering services to customers with a short service downtime among servers, thus become the number of active server decrease and moreover, the energy consumption of whole cloud data center will be decreased.

Server utilization and energy consumption can be improved by not allowing new workloads to be accepted in situations where temperature and energy consumption of server increase dramatically and execute them on other servers. On the other hand, because of variability of workloads sometimes virtual machines can't get enough resources from host (because of host resource limitation) and need to move virtual machines from a particular server to other servers by using live virtual machine with the aim to energy efficiency and improving temperature.

To improve dynamic virtual machine consolidation and because of the complexity of the problem, we divided into 4 subproblem based on [8].

1. Server overload detection
2. Server under load detection
3. Virtual machine selection
4. Virtual machine placement

The rest of the paper is organized as follows: In Sect. 2, related work is discussed. Section 3 presents the models and metrics used in the paper, including the energy consumption model, virtual machine migration cost model, the negotiated SLA, and performance metric. After that, the virtual machine consolidation techniques are introduced in Sect. 4. Simulation results and evaluation are shown in Sect. 5. Finally, Sect. 6 presents conclusions and future work.

2 Related work

In the literature, there has been great progress and studies in data center efficiency and utilization over the past ten years by students, researchers, and companies to improve the energy efficiency of cloud data centers. Cao and Dong [9] has done lots of research to gain a relationship between performance and energy. They attempt to balance between energy and performance by redesigning a new energy-aware heuristic framework for virtual machine consolidation. In the framework, they define an algorithm for service-level agreement violation to determine when a server is overloaded by using three metrics: SLA violation, minimum power consumption, and maximum utilization. The aim of this work is to improve the power consumption while maintaining the quality of service. Deng et al. in [10] proposed a threshold in server consolidation based on multidimensional resource utilizations such as CPU, memory, and disk. When server load reaches the threshold, server consolidation algorithms run and balance servers to prevent server under load or overload. In other words, they define a trigger, based on server parameters to call server consolidating algorithm and run it. Virtual machine consolidation has been considered by Younge et al. [11] focusing on energy efficiency of data centers. The proposed framework focuses on minimizing the number of active servers to reduce energy consumption by the server consolidation; however, the authors try to consolidate virtual machine to decrease the number of active servers, but it ignored other factors such as temperature, workload, server type, and network traffic. They also presented a new virtual machine system image to reduce the size of the virtual machine with the aim of reducing the time of virtual machine start-up and virtual machine migrating.

Arianyan et al. in [12] concentrated on consolidation problem to improve resource utilization and efficiency to cause energy reduction in cloud data centers. They introduced new resource management technique according to three parameters: energy consumption, live migration, and SLA violation. They divide the whole problem into 2 subproblems, first, find overloaded hosts, and second, find a suitable virtual machine for migration by using multi-criteria decision-making techniques.

Beloglazov et al. in [13] introduced a dynamic server consolidation framework. The framework uses an upper and lower thresholds based on CPU utilization. The framework tried to hold server utilization between upper and lower thresholds and virtual machines from overloaded and underloaded servers are submitted to a modified best fit decreasing algorithm to decide on virtual machine placement in a way of energy reduction and minimize the number of SLA violations. They proposed new dynamic threshold algorithm that periodically set upper and lower threshold based on recent resource utilization of server with aim of helping tradeoff between performance and energy consumption [8]. Such upper and lower threshold algorithms are done by Yang et al. in [14]. However, they mainly focus on energy reduction without attention to performance and migration cost.

Li et al. in [15] introduced new virtual machine consolidation mechanism to plan for virtual machine consolidation with regard to the decisions of IT administrators. To participate decision of administrators, they used a language to describe manager's preferences and metrics to evaluate the satisfaction degree to map virtual machines on an optimum server with the goal to reduce resource wastage on servers. To avoid SLA violation and guarantee quality of services, the virtual machines that cause perform degrading should be moved to other idle servers.

In [16], Hwang and Pedram proposed a heuristic solution to solve the problem efficiently in server consolidation. The proposed solution considered correlation and multiple resource type to prevent bottlenecks and random workload; it also includes two resource managers, a global manager for assigning virtual machines to a cluster and a local manager to run virtual machines on servers.

In [17], Mastroianni et al. presented a solution for the consolidation of virtual machines based on Bernoulli trials. The solutions have two main features which include a self-organizing and adaptive method that makes it efficient for using in the large data center. By using Bernoulli trials and local information, the server decides to be or not available for running applications. One of the best benefits of this solution is balancing CPU-bound and memory-bound applications.

Farahnakian et al. to address virtual machines placement proposed an online metaheuristic algorithm based on Ant Colony System. The algorithm determines a near-optimal solution based on a specified objective function. As a result, they improve energy consumption while maintaining the required performance levels in a cloud data center [18].

Beloglazov and Buyya [19] proposed a method for host overload detection of virtual machine consolidation technique based on Markov chain and a control algorithm for both known stationary and unknown nonstationary workloads. Also for the management of unknown nonstationary workloads used multi-size sliding windows in workload estimation.

In [20], Karthik et al. considered the problem of virtual machine consolidation focusing on resource profiling and virtual machine placement. Resource profiling is responsible for understanding the trend of resource demands for new jobs and handling jobs with the minimum virtual machine. They implemented 3D Bin Completing algorithms for virtual machine consolidation using parameters such as CPU, Memory, and IO as the dimensional domain instead of focusing on CPU component of the server.

Zola and Kassler [21] proposed a novel method for the problem of energy-efficient virtual machine consolidation based on the theory of robust. The approach allows a data center operator to specify this risk aversions. Data center operators can have a choice of selecting more robust solutions at a higher energy or more opportunistic solutions that have a higher probability of constraint violation but at a lower energy cost.

Live migration brings us the movement of virtual machines with a short service downtime among servers, during migration, it entails CPU overhead, and consequently CPU overhead degrades performance. Wu and Kassler [22] attempt to preserve energy using virtual machine consolidation in cloud data center while keeping most of the virtual machine unmoved, because of virtual machine performance degradation during the migration period. They proposed an improved genetic algorithm based on virtual machine consolidation and reduce migration cost.

Raising equipment temperature due to overutilization becomes ICT resource facing with heat problem. Like any other equipment data center resources are susceptible to damage from heat such as educed system reliability and availability and also cause to increase CO2. In [23], Lee et al. have performed a comprehensive research on thermal-aware consolidation. They use a thermal threshold to control resource consolidation and improve the efficiency of heat extraction by identifying and comparing the thermal sensitivity of servers with an inlet air temperature.

Wang et al. [24] proposed thermal-aware virtual machine allocation algorithm with regard to the temperature of processors. They implemented a lumped RC thermal model to describe the thermal behavior of the processors. The proposed mechanism selects and allocates virtual machines to servers based on server's temperature.

3 Evaluation models and metrics

3.1 Power model

In this paper, we define our power consumption as a CPU utilization function based on Anton defined in [13]. The function is shown in Eq. (1):

$$P(u) = K \times P_{\max} + (1 - K) \times P_{\max} \times u \tag{1}$$

where $P_{\max}$ is the maximum power of server in the running state and experienced the CPU utilization of 100%; k is the percentage of power consumed by an idle server; and the CPU utilization is denoted by u. Because the utilization of a CPU changes over time, we defined it as a function $u(t)$ of time. Therefore, total energy consumption can be obtained by Eq. (2):

$$E = \int_{t}^{\infty} P(u(t))\mathrm{d}t \tag{2}$$

According to this function, the energy consumption of the server is determined by the CPU utilization. Thus, to reduce the whole energy consumption, we take into account the CPU utilization in consolidation technique.

3.2 Live migration cost

Live migration of virtual machines allows moving virtual machines between servers with a short service downtime among servers. However, live migration moves virtual machines without any noticeable influence from the end user view but has a bad influence on the performance of applications. A recent research on [25] shows the virtual machines migration influences on both the source and destination performance. Thus, the number of migrations should be reduced. For the sake of improving performance, we use live migration cost model that was proposed in [8] to select suitable virtual machine for migration to minimizing the number of live migration. The author in [8] shows that a single live migration can cause performance degradation and can be estimated about 10% cpu overhead, thus each live migration may cause some SLA violation. Our goal is to reduce the number of live migrations, so we use the migration time and performance degradation formula experienced by virtual machine j based on [11] in Eqs. (3) and (4):

$$U_{\mathrm{d}_j} = 0 \cdot 1 \int_{t_0}^{t_0+T_{\mathrm{m}_j}} u_j(t)\mathrm{d}t \tag{3}$$

$$T_{\mathrm{m}_j} = \frac{M_j}{B_j} \tag{4}$$

where U_{d_j} is the total performance degradation by virtual machine j, t_0 is the time when the migration starts, T_{m_j} is the time taken to complete the migration, $u_j(t)$ is the CPU utilization by virtual machine j, M_j is the amount of memory used by virtual machine j, and B_j is the available network bandwidth.

3.3 Service-level agreement violation metrics

In a cloud computing environment, there are many users that request for resources and services at the same time. While the users' request competes with each other to obtain resources, users want to ensure their services provided based on service-level agreement, such as minimum bandwidth, maximum response time, and maximum downtime defining a unique metric to evaluate service-level agreement that independent of workload is necessary for the evaluation of our work with others. The author in [8] defined SLA violation using two metrics for evaluating in IaaS model:

1. The percentage of time which servers experienced the cpu utilization of 100%, SLA violation Time per Active Host (SLATAH). The function is shown in Eq. (5)
2. The overall performance degradation by virtual machines due to live migrations, also known as performance degradation due to migrations (PDM). The function is shown in Eq. (6)

$$\text{SLATAH} = \frac{1}{N}\sum_{i=1}^{N}\frac{T_{\mathrm{s}_i}}{T_{\mathrm{a}_i}} \tag{5}$$

$$\text{PDM} = \frac{1}{Q}\sum_{j=1}^{Q}\frac{C_{\mathrm{d}_j}}{C_{\mathrm{r}_j}} \tag{6}$$

where N is the number of servers; T_{s_i} is the total time during which the server i has experienced the utilization of 100%, leading to an SLA violation; T_{a_i} is the total of the server i being in the active state; M is the number of virtual machines; C_{d_j} is the estimate of the performance degradation of the virtual machine j caused by migrations; C_{r_j} is the total CPU capacity requested by the virtual machine j during its lifetime. In our experiments, we set C_{d_j} as 10% of the CPU utilization in MIPS during all migrations of the virtual machine $j$based on [8]. Beloglazov and Buyya in [8] proposed a combined metric that involves both performance degradation due to server overloading and due to virtual machine live migrations. SLA violation (SLAV) is calculated as shown in Eq. (7).

$$\text{SLAV} = \text{SLATAH} \times \text{PDM} \tag{7}$$

3.4 Performance metrics

The author in [8] defines a new metric by multiplying energy consumption and SLA violations to compare the efficiency of the algorithms with others. They denote energy and SLA violations (ESV) as shown in Eq. (8)

$$\text{ESV} = E \times \text{SLAV}. \tag{8}$$

4 Energy-aware virtual machine consolidation

Energy consumption by the server in data centers is mostly determined by the CPU and memory, in which CPU is the main power consumers in servers, so concentrating on improving CPU utilization benefits in both utilization and energy consumption. Studies show that on the average, an idle server consumes a significant amount of the power proportional servers running at full CPU speed [26]. This fact led to the idea of consolidating virtual machine in a fewer server and hold these servers at full utilization. On the other hand, consolidating virtual machines on the high-performance

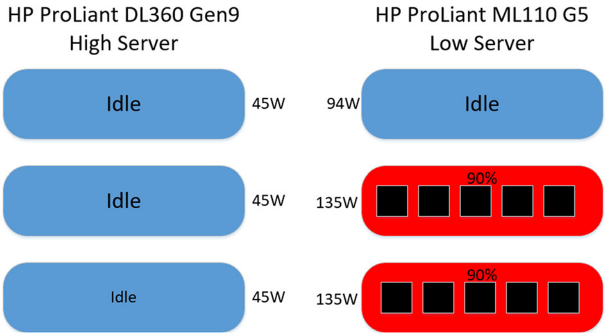

Fig. 1 Virtual machine consolidation in low performance server

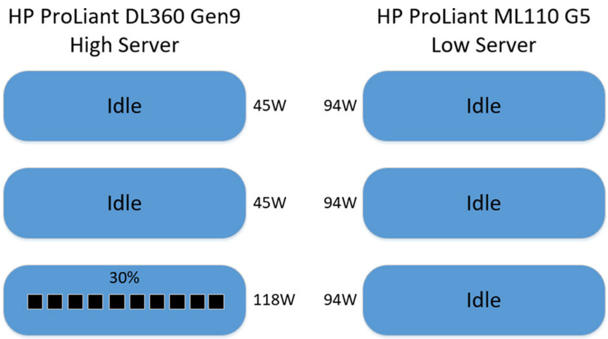

Fig. 2 Virtual machine consolidation in high performance server

server or low-performance is one of the new challenges that we cover in this paper. It should be noted high-performance servers produce less heat in comparison with low-performance servers while having more workload and consuming less power.

As it can be seen from Fig. 1 virtual machine consolidation is done; therefore, the least servers are active and energy consumption of whole servers is equal to 499w, but the temperature of these servers is not normal and causes the heat problem.

Now we wanted to consolidate those virtual machines in high-performance server instead of low-performance server. As it can be seen from Fig. 2, in addition to server consolidation, it also causes servers to be in normal state from temperature point of view. Also, energy consumption of whole servers equals 490. The energy consumption of two methods is close together, but in the second method it eliminates the heat problem of low-performance servers. As described earlier, virtual machine dynamic consolidation packs a number of virtual machines on a fewer number of servers to optimize resource utilization and reduce power consumption. As mentioned before because virtual machine consolidation is complex, we divided the problem into four steps:

1. Server overload detection
2. Server under load detection
3. Virtual machine selection
4. Virtual machine placement

In the following, we focus on each step thoroughly.

4.1 Server overload detection

The first part of server consolidation determines whether a server is overloaded or not, and when to migrate some virtual machines from the overloaded server. We recognize an upper bound threshold for server utilization; to find upper threshold, we examine all servers in the data center and find the best server based on a server that has the maximum number of operations per second (OPS) per watt. This server is known as the optimum, and this optimum server helps us to determine the upper threshold for other servers. As mentioned earlier, about 10% of extra CPU utilization need for live migration, thus cannot put server utilization on 100% because of SLA violation, hardware failure, and life span of the server. After simulating on many server and with attention to live migration overhead (about 10% cpu overhead), we select the upper threshold for the optimum server as 90% CPU utilization. Thus, we set 90% CPU utilization as an upper threshold for this server and then proceeded to calculate the temperature of the server in 90% utilization based on [26]. After the temperature of server obtained at 90% CPU utilization, we pick it as temperature threshold. In the next step named it as $\text{Threshold}_{\text{temperature}}$ and do virtual machine consolidation based on this threshold on another server, in other words, we do a thermal-aware consolidation on other servers. The benefits of this algorithm are to control server performance and server temperature together. Thus, for other servers, if the temperature crosses Threshold temperature, it detects that server as overload. The general algorithm of server overload detection is shown in the Algorithm 1. The complexity of the server overload detection is $n + n = O(n)$, where n is the number of servers.

Algorithm 1: Server overload detection

```
Input: Server List
Output: Overload Server List
N = 1
//PoE stands for Performance Over Energy
PoE_maximum = 0
For each Server in Server List do {
      PoE_N ← P/E // Performance Over Energy
      Increase N
      If PoE_N > PoE_maximum {
            PoE_maximum = PoE_N
            Optimum Server = Server
            }
}
Threshold_temperature = Temperature of Optimum Server_Utilization 90%
For each Server in Server List do {
Threshold = Get Temperature of Server
      If Threshold > Threshold_temperature {
            Server is Overload
            Add Server to Overload Server List
            }
}
Return Overload Server List
```

4.2 Server under load detection

In opposite of server overload detection, we need to determine whether a server is under load to migrate all virtual machines to other servers. The algorithm collects virtual machines from underload servers to be mapped in a way to minimize the number of active servers and switch the remaining idle host into sleep mode and eliminate energy consumption of idle servers. The proposed algorithm categorized server based on type and characteristics such as CPU, memory, bandwidth, and power and put every server in one of the following categories:
Low category servers that consume most energy consumption against other servers in the data center while doing lowest operation per second.
Medium category these servers are better than low category but not as high category
High category the server that consumes minimum energy consumption against other servers in the data center while doing maximum operation per second.

We proposed a server under load detection algorithm to minimize the number of low category servers. Primarily try to determine the server that provides the most increase in power consumption in low category, if do not find any server in this category, consequently continue to explore in the medium and high category. Thus, if we move the virtual machine from low category server to another server, in addition to improve the energy consumption of servers and decrease the number of active servers, it also aims to minimize the number of low category servers.

The more the number of low category server active, the more probability of overload occurrence which can lead to more resource shortages and consequent increase in migrations. The general algorithm of the server under load detection is shown in Algorithm 2. N is the number of server type and greater N means high- performance

Algorithm 2: Server under load detection

```
Input: Server List,
Output: Under loaded Server
Categorized Server Based on Type
N = Get Number of Server Type
For 1 to N do {
      Minimum_N = 1
      For each Server in Type N do {
            Server Utilization ← Get Server Utilization
            If Server Utilization < Minimum_N {
                  Under load Server in Type N = Server
                  Minimum_N = Under load Server in This Category
            }
      }
}
For 0 to N {
      If there is Under Load Server in Type N{
      Under load Server = Under load Server in Type N
      Exit for loop
      }
}
Return Under load Server
```

server with less energy consumption. The complexity of algorithm is $nm+n = O(nm)$, where m is the number of servers in specific type and n is a constant that indicates the number of server type; in this paper, we selected n as 3 (low, medium, high) values, so the complexity equal to $3m + 3 = O(m)$.

4.3 Virtual machine selection

There are more than one virtual machines run on a server; thus, it needs to consider how to choose a migratable virtual machine when a server is overloaded. Different parameters involve to select the virtual machine, and at the moment, there are four types of virtual machine selection policies: maximum correlation (MC), minimum migration time (MMT), minimum utilization (MU), and random selection (RS) [19]. MC migrates a virtual machine that has the maximum correlation coefficient compared to the other virtual machines on the same server. The MMT migrates those virtual machines that will take the least time to move. MU selects a virtual machine with the lowest utilization. And, RS migrates virtual machines randomly without any rules. We propose a novel virtual machine selection algorithm which calculates the deviation between utilization of overloaded server and its threshold and find the virtual machine in which virtual machine utilization is close to the deviation. we named it maximum fit (MF) because we wanted to find maximum fit virtual machines for migration; in other words, we want to get a server back to the normal state with the minimum number of live migration. We find the virtual machine by binary search in all the virtual machines on overload server to decrease the number of migrations to prevent performance degradation due to live migration. The complexity of algorithm is $\log n$, where n is the number of virtual machine in overloaded server.

4.4 Virtual machine placement

After selecting virtual machines that need to be migrated, determining servers that host those virtual machines is the last steps of virtual machine consolidation. One method used in [8] is the Power-Aware Best Fit Decreasing (PABFD), it allocates each virtual machine to a server that provides the least increase in power consumption due to this allocation. We mentioned in Sect 4.2. A number of server type consumes more energy consumption against the other while doing less operation per second and we attempt to select such servers to improves energy consumption of servers, decrease the number of active servers and aim to decrease the number of low category servers, but here is vice versa. We propose a new algorithm to allocate virtual machines to the server that select the server within high category servers to preserve high category server active and low category servers inactive. The core idea behind this is the low category servers consume more energy consumption per operation against the other. Thus, if we placed virtual machine to this server, it improves the energy consumption of server and aims to select a high category server in advance. The goal of this algorithm is to allocate virtual machines to the server that provides the least increase and has more resource capacity for virtual machine placement in future. The core idea behind this is the more resource capacity, the less probability of overload occurrence which can lead to fewer resource shortages and consequent decrease in migrations.

We have done our test on different servers with different characteristics to examine our idea, and the result shows that if we select a high category server that provides the least increase in power consumption due to this allocation, it decreases live migration and accordingly performance degradation due to live migration and SLA violation in comparison with selecting low category server. The general algorithm of the server under load detection is shown in Algorithm 3. N is the number of server category and greater N means the high-performance server with less energy consumption. The complexity of algorithm is $nm + n = O(nm)$, where m is the number of servers in specific type and n is a constant that indicates the number of server type; in this paper, we selected n as 3 (low, medium, high) values, so the complexity equal to $3m + 3 = O(m)$.

Algorithm 3: Maximum fit (MF)

```
Input: Overload Server
Output: Migratable Virtual Machines
Virtual Machines List ← Get Virtual Machines of Server
N← Get Number of Virtual Machine on Server
Threshold_Upper ←Find Upper threshold of Server
Sort virtual machines in Host in Ascending order based on
utilization
Deviation = Server Utilization – Threshold_Upper
First = 0
Last = N – 1
Middle = (Last + First)/2
If Virtual Machine_middle = Deviation {
    Migratable Virtual Machines ← Virtual Machine_middle
    }
while (first < last)
{
    middle = (first + last)/2;
    if (Virtual Machine_middle = Deviation) {
        Migratable Virtual Machines ← Virtual Machine_middle
        Exit While
        }
    if (Virtual Machine_middle < Deviation) {
        first = middle + 1;
        }
    else {
        last = middle – 1;
        }
    middle = (first + last)/2;
            }
If Migratable Virtual Machines is null {
    Migratable Virtual Machines ← Virtual Machine_first
    }
    Return Migratable Virtual Machines
```

5 Performance analysis

5.1 Simulation experiment

Advanced modeling and simulation methodologies are essential aspects of performance evaluation that precedes the costly prototyping activates required for complex, large-scale computing, control, and communication systems. Simulation environments allow evaluation of different kinds of resource leasing scenarios under varying load and pricing distributions. The CloudSim toolkit [27] has been chosen as a simulation platform; it is a modern simulation framework for cloud computing environments. It allows the modeling of virtualized environments, supporting on-demand resource provisioning and their management; it also has been extended to enable energy-aware simulations. Apart from the energy consumption modeling and accounting, the ability to simulate service applications with dynamic workloads has been incorporated. The implemented extensions have been included in the 3.0.3 version of the CloudSim toolkit. We have simulated a data center that comprises 300 heterogeneous servers, hundreds of them are HP ProLiant ML110 G5 servers, hundreds of them are HP ProLiant DL360 G7, and the rest hundreds are of HP ProLiant ML110 G9 servers. The characteristics of the servers are given in Table 1 based on [26].

The characteristics of the virtual machine types are based on Amazon EC2 instance types [28] with the only exception that all the virtual machines are single core because the workload data used for the simulations come from single-core virtual machines. Initially, the virtual machines are allocated according to the resource requirements defined by the virtual machine types. However, during the lifetime, virtual machines utilize less resource according to the workload data, creating opportunities for dynamic virtual machines consolidation. The characteristics of the virtual machines types are given in Table 2.

Table 1 Specification of servers

Server	HP ProLiant ML110 G5	HP ProLiant DL360 G7	ProLiant DL360 Gen9
Processor (Mips)	2660	3067	2300
Number of core	2	12	36
Memory (GB)	4	16	64
Network BW (GB/s)	1	1	1

Table 2 Specification of virtual machines

Virtual machines	Large	Medium	Small	Micro	Nano
Processor (Mips)	2000	1000	1000	500	250
Memory (MB)	2048	2048	1024	1024	512
Network bandwidth (GB/s)	1	1	1	1	1

Table 3 Workload data characteristics

Date	Number of virtual machines
2011/03/03	1052
2011/03/06	898
2011/03/09	1061
2011/03/22	1516
2011/03/25	1078
2011/04/03	1463
2011/04/09	1358
2011/04/11	1233
2011/04/12	1054
2011/04/20	1033

In order to make the results more realistic, it is important to conduct experiments using workload traces from a real system. We have used data that provided as a part of the CoMon project, a monitoring infrastructure for Planet Lab [29]. We have used the data on the CPU utilization by more than a thousand virtual machines from servers located at more than 500 places around the world. The interval of utilization measurements is 5 min. We have randomly chosen 10 days from the workload traces collected during March and April 2011. The characteristics of the data for each day are shown in Table 3.

5.2 Performance metrics

There are many metrics to measure the efficiency and superiority of various algorithms. The first metrics for evaluation is energy consumption of servers which can be calculated according to the energy model mentioned in Sect. 3. In this model, the parameter $\mathbf{P_{max}}$ and the value of coefficient **k** are based on servers are shown in Table 4.

Another metric is SLA violation percentage, which is defined in Sect. 3. The third metric is the total number of virtual machine migrations in the data center. The fourth metric is performance metric which is defined in Sect. 3, and the last metric calculates the temperature of a server. Using workload data, we compare our work with the algorithms IQR and LR and virtual machine selection policies MC and MMT according to Beloglazov and Buyya [8]; these algorithms are the best algorithms among the others mentioned before. In this paper, our proposed algorithm, known as “DthMf,” stands Dynamic Threshold Maximum Fit.

Table 4 $\mathrm{P_{max}}$ and coefficient k

Server	HP ProLiant ML110 G5	HP ProLiant DL360 G7	ProLiant DL360 Gen9
$P_{\max}$ (W)	135	216	276
K (%)	69.40	25.74	16.3

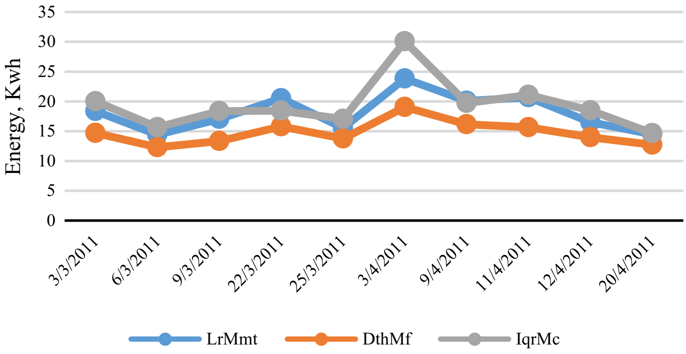

Fig. 3 Energy consumption

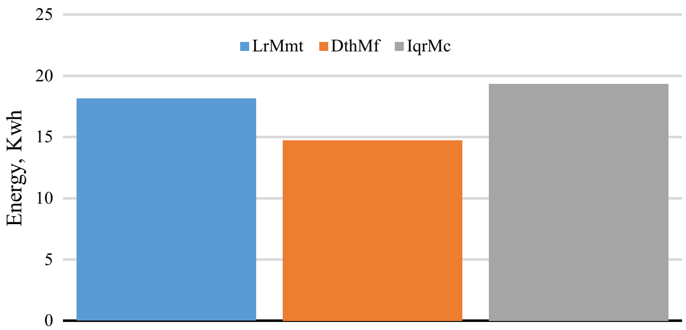

Fig. 4 Average energy consumption

5.3 Simulation result

As it is quite clear from the chart in Fig. 3 that our proposed algorithm "DthMf" has the lowest energy consumption among the other algorithms in all ten days, and the reason is the algorithm investigates to determine the upper threshold for the servers based on server category. The core idea behind the algorithms is setting an upper threshold for each server based on server characteristics, for example for the high category server, set upper threshold as 90% CPU utilization while for a medium category server set 80% CPU utilization. Figure 4 shows the average energy consumption in these ten days. The lowest energy consumption belongs to DthMf (14.74 kW) in comparison with IqrMc (19.34 kWh) and LrMmt (18.14 kW).

One of the most important factors that affect energy consumption and SLA violation is the number of live migration. The chart in Fig. 5 shows comparison between the DthMf, IqrMc, and LrMmt from the number of live migration point of view. It can be seen from the chart that DthMf has the minimum number of migrations in all ten days. This is due to virtual machine selection algorithm attempt to select the maximum fit virtual machine to return server status to normal state with minimum migration as possible. Figure 4 shows the average number of migrations in these ten days the

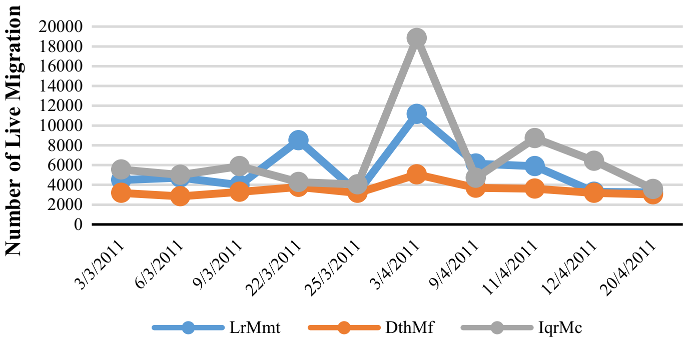

Fig. 5 Number of live migration

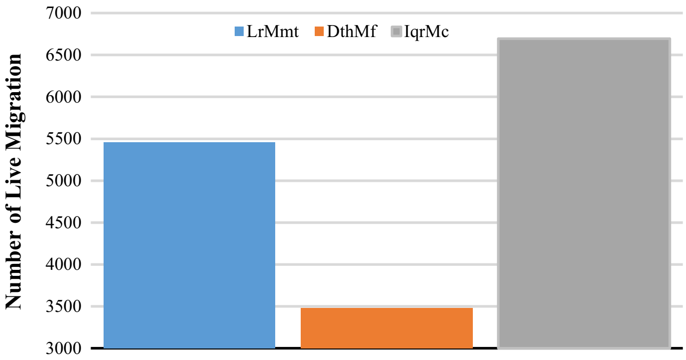

Fig. 6 Average number of live migration

minimum number of migrations belongs to DthMf (3481) in comparison with IqrMc (6696) and LrMmt (5460) as shown in the chart of Fig. 6.

Before discussing the result analysis of SLA violation, once again we refer to Sect. 3 where we show that SLA obtained by multiplying SLA violation Time per Active Host and Performance Degradation due to migrations.

$$\text{SLAV} = \text{SLATAH} \times \text{PDM} \tag{9}$$

Therefore, with reducing one of the variables (SLATAH or PDM) in multiple, the final answer (SLAV) also reduced. As mentioned earlier, DthMf reduced the number of live migration, less number of live migration means that less number of performance degradation due to migration (PDM) occurred; therefore, the SLAV reduced. Figure 7 shows DthMf improves SLA violation in all ten days in comparison with others. Figure 8 shows the average SLA violation in these ten days and it is quite clear from the chart in figure that the minimum SLA violation in ten days belongs to DthMf (0.681×10^{-5}) in comparison with IqrMc (1.45×10^{-5}) and LrMmt (1.15×10^{-5}).

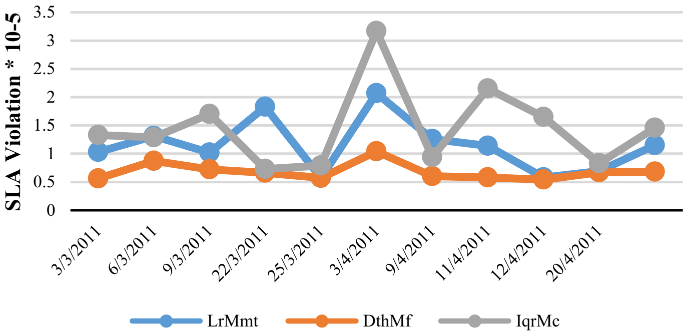

Fig. 7 SLA violation

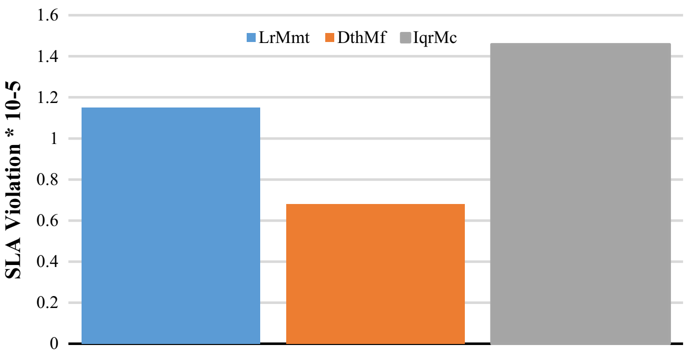

Fig. 8 Average SLA violation

We refer to Sect. 3 where we show that ESV was obtained by multiplying energy consumption and SLA violation.

$$\mathrm{ESV} = E \times \mathrm{SLAV} \tag{10}$$

If we decrease SLA violation to zero, it is obvious that the ESV becomes zero; so keep in mind that as much as ESV closer to zero it is better. It is quite clear from the chart in Fig. 9 DthMf is more efficient in comparison with IqrMc and LrMmt; therefore, with reducing SLA violation and energy consumption, the ESV also reduced. Figure 10 shows the average ESV in these ten days, and the minimum ESV belongs to DthMf (10.14×10^{-5}) in comparison with IqrMc (30.67×10^{-5}) and LrMmt (21.92×10^{-5}).

Virtual machine consolidation attempts to consolidate virtual machines on the server until reach upper threshold. On the other hand, there is a logical relationship between utilization and temperature of the servers based on [26]. For example, in HP ProLiant ML110 G5 when server utilization reaches to 90%, the temperature rises to about 22.2 °C as it is shown in Fig. 11. Therefore, improving server utilization with respect to temperature helps to control energy consumption and heat problems.

As it can see from the chart in Fig. 12, the upper threshold is different for each server and each algorithm. In algorithm LrMmt for all servers, the upper threshold

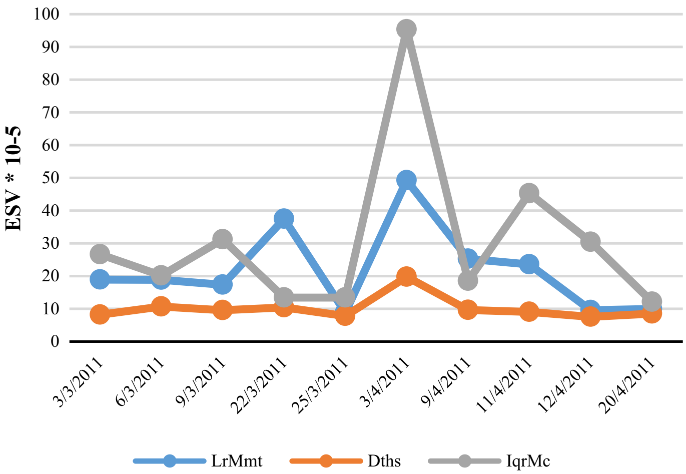

Fig. 9 ESV

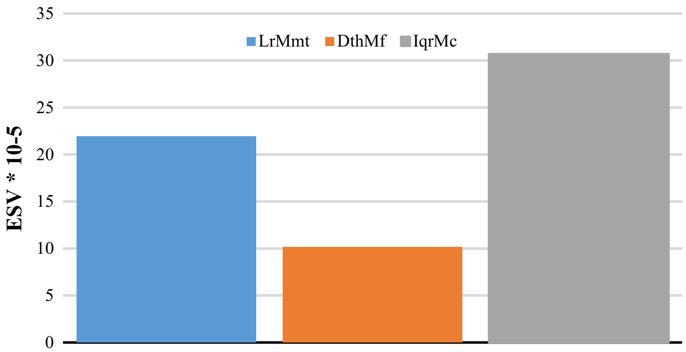

Fig. 10 Average ESV

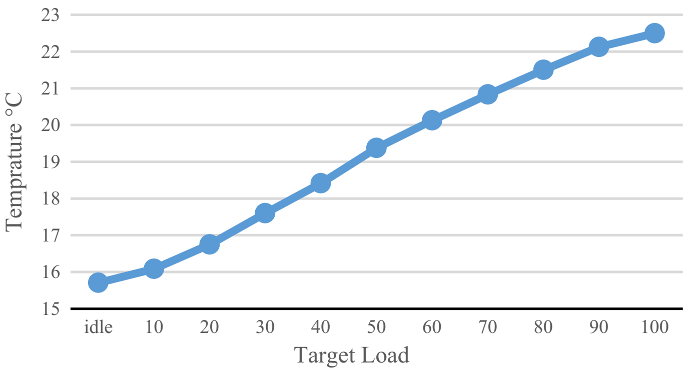

Fig. 11 HP ProLiant ML110 G5 server

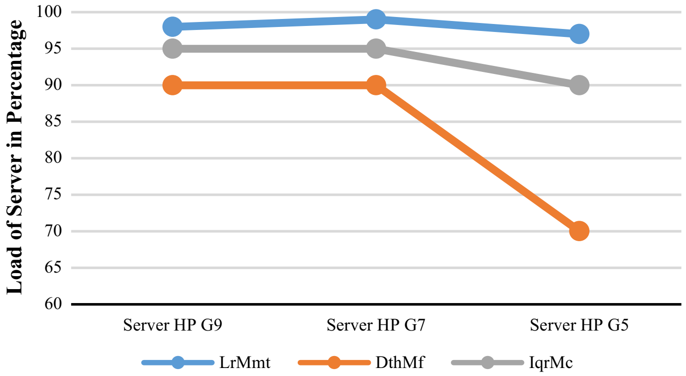

Fig. 12 Server thresholds

is about 98% of CPU utilization. Algorithm 4 for HP G9 and HP G7 servers sets the upper threshold to about 95% of CPU utilization and for HP G5 server sets the upper threshold to about 90% of CPU utilization. Finally, our algorithm has set upper threshold for HP G9 and G7 servers about 90% of CPU utilization and for HP G5 server upper threshold of about 70% of CPU utilization.

```
Algorithm 4: Virtual machine placement

Input: Server List,
Output: Suitable Server
Categorized Server Based on Type
N = Get Number of Server Type
For 1 to N do {
      Minimum_N = 1
      For each Server in Type N do {
            Server Utilization ← Get Server Utilization
            If Server Utilization < Minimum_N {
                  Under load Server in Type N = Server
                  Minimum_N = Under load Server in This Category
            }
      }
}
For N to 0 {
      If there is Load Server in Type N{
      Suitable Server = Under load Server in Type N
      Exit for loop
      }
}
Return Suitable Server
```

After that, we calculated the temperature of servers which has been produced in the upper threshold based on [26]. The results are shown in Fig. 13; in the LrMmt algorithm for HP G9 and G7 servers, the temperature is about 21.5 °C and for HP G5 server temperature is about 22.5 °C in situations in which server load is close to upper

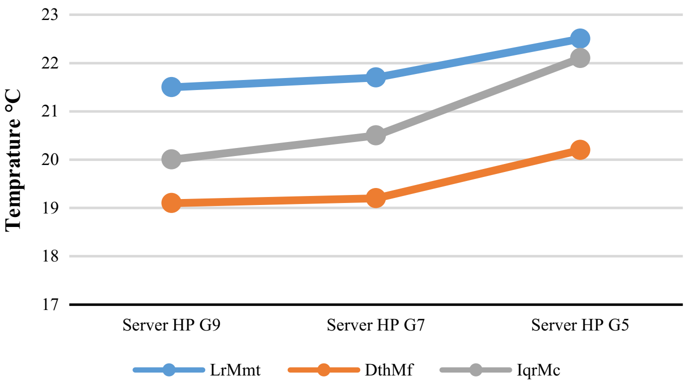

Fig. 13 Server temperature

threshold. In IqrMc algorithm for HP G9 server, the temperature is about 20 °C, for HP G7 server the temperature is about 20.5 °C, and for HP G5 server temperature is about 22 °C when server load is close to upper threshold. Finally, in DthMf for HP G9 and G7 server, temperature is about 20.5 °C and for HP G5 server temperature is about 20 °C in situations in which server load is close to upper threshold. Thus, DthMf in addition to improving improve energy consumption, decreasing the number of migration and SLA violation, also aims to decrease the temperature of the servers.

6 Conclusion and future work

To maximize cloud provider's benefits, it is necessary to reduce the energy consumption without any side effect or degradation in service-level agreement in cloud services. For energy efficiency, resource management using virtual machines consolidation plays an effective role in cloud computing. Note that it is very critical to manage the energy consumption using virtual machine consolidation while supporting the minimum requirement for service quality. In this paper, we have improved the virtual machine's consolidation algorithm that supports the proper algorithms for server overload detection, server underload detection, virtual machine selection, and virtual machines placement. We have evaluated the proposed algorithm through extensive simulations with a large-scale experiment setup using workload traces from more than a thousand Planet Lab virtual machines. Results of the experiments have shown that the proposed DthMf algorithm significantly improves in metrics energy consumption, SLA violation, and number of live migration than other dynamic virtual machine consolidation algorithms. As future work, we intended to extend these algorithms by taking into account network elements such as routers, switch, and bandwidth to consider communication in virtual machine consolidation and other parameters such as memory and storage to consider more parameters in virtual machine consolidation. To evaluate the proposed system in a real cloud infrastructure, we will run our proposed algorithms on Open Stack and Open Nebula.

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